

The Boolean Kernel Basis and Its Low-Degree Filtration over Arbitrary Fields

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Abstract

For $N = 2^m$ and an arbitrary field $\mathbb{F}$, we study the family of kernel polynomials $K_y(X) = \prod_{i=0}^{m-1} (X^{2^i} + y_i)$ indexed by $y \in \{0, 1\}^m$. We show that this family is a basis of $\mathbb{F}[X]_{<N}$ in which *every* basis element has degree exactly $N - 1$: unlike degree-graded bases such as the monomial basis or the Lin–Chung–Han novel polynomial basis, no element of the kernel basis has low degree. We then characterize the ordinary degree filtration of $\mathbb{F}[X]_{<N}$ directly in kernel coordinates. Writing each index as $y = (x, h)$, with x ranging over k low coordinates and h over the remaining $q = m - k$ high coordinates, we prove that a polynomial U has degree less than 2^k if and only if its kernel-coefficient table satisfies $\lambda_{(x,h)} = (-1)^{q-\text{wt}(h)} \mu_x$ for a single function μ on the low coordinates – an alternating sign condition that is dual, under the zeta/Möbius transform on the Boolean lattice, to the trivial vanishing of high monomial coefficients. We give both directions of this equivalence, an explicit collapse formula recovering U from μ , and the resulting subspace $CH_{m,k} \subset \mathbb{F}^{\{0,1\}^m}$ of dimension 2^k . All results hold over arbitrary fields, with no characteristic or subspace assumption. A committed Rust implementation provides a reproducible computational cross-check of the coefficient transform and both directions of the filtration theorem, while a Lean 4 formalization is in progress and is not presented as a completed machine-checked proof.

Keywords: finite fields, polynomial bases, kernel polynomials, Boolean lattice, Möbius transform, degree filtration

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1. Introduction

The relationship between a polynomial's degree and the coefficients that describe it depends entirely on which basis those coefficients are taken in. In the monomial basis $\{1, X, X^2, \dots\}$ this relationship is definitional: U has degree less than 2^k exactly when its coefficients on monomials of degree 2^k and above vanish. Every *degree-graded* basis of $\mathbb{F}[X]_{<N}$ – one in which the j -th basis element has degree j – inherits the same picture for free: sparsity of the coefficient table above index $2^k - 1$ is the degree bound.

This paper studies a basis for which that picture fails completely. Index $\mathbb{F}[X]_{<N}$, $N = 2^m$, by Boolean vectors $y \in \{0, 1\}^m$ via the kernel polynomials

$$K_y(X) = \prod_{i=0}^{m-1} (X^{2^i} + y_i).$$

Theorem 4.5 shows $\{K_y\}$ is a basis of $\mathbb{F}[X]_{<N}$ over an arbitrary field $\mathbb{F}$, with no assumption on characteristic. But every single kernel polynomial has degree exactly $N - 1$ (Proposition 4.2): the basis is maximally degenerate with respect to degree, and no sub-collection of its elements spans a proper degree-filtered subspace. If a polynomial U has small degree, that fact cannot show up as *which* kernel coefficients are zero, because none of the individual basis elements are themselves low-degree.

Our main result, Theorem 5.2, identifies what a degree bound looks like instead. Splitting a kernel index $y = (x, h)$ into k low coordinates and $q = m - k$ high coordinates, U has degree less than 2^k if and only if its kernel coefficients satisfy

$$\lambda_{(x,h)} = (-1)^{q-\text{wt}(h)} \mu_x$$

for a single function μ of the low coordinates alone – an alternating sign condition across every fiber of the high coordinates, rather than a support condition. This is the Boolean-lattice Möbius dual of the trivial monomial statement: where the monomial basis reads off degree as which coefficients vanish, the kernel basis reads off degree as a rigid sign pattern relating the coefficients that remain.

Bases indexed by $\{0, 1\}^m$ and built from a per-coordinate recursive product are not new in themselves: the Lin–Chung–Han “novel polynomial basis” [1] plays a comparable role in additive Fourier transforms over characteristic-two fields, and underlies the folding step of recent FRI-based polynomial commitment schemes [2, 3]. Section 2 makes the comparison precise; in short, the Lin–Chung–Han basis is degree-graded and tied to a chosen subspace structure over a field of characteristic two, so its interaction with degree is the ordinary sparsity picture. The kernel basis studied here is field-agnostic and maximally degenerate in degree instead, which is exactly what makes Theorem 5.2 necessary rather than immediate. We view the degree-halving behavior of an FRI-style folding step as a natural setting where a result like Theorem 5.2 may be put to use. This paper is self-contained and establishes the underlying algebra on its own terms.

Contributions.

- We prove that the kernel polynomials $\{K_y : y \in \{0, 1\}^m\}$ form a basis of $\mathbb{F}[X]_{<N}$ over an arbitrary field, and that every element of this basis has degree exactly $N - 1$ (Theorem 4.5, Proposition 4.2).
- We give the exact change of basis relating kernel and monomial coordinates, as a Boolean zeta transform composed with a complement permutation, invertible by Boolean Möbius inversion in $O(N \log N)$ operations (Section 4.4).
- We characterize the ordinary degree filtration of $\mathbb{F}[X]_{<N}$ exactly in kernel coordinates, as an alternating-sign condition rather than a sparsity condition (Theorem 5.2), together with an explicit collapse formula (Proposition 5.3) and the resulting 2^k -dimensional subspace $CH_{m,k}$ (Corollary 5.5).
- We provide a committed Rust cross-check of the coefficient formula, zeta/Möbius transforms, and both directions of the filtration theorem over the Goldilocks prime field, and report separately on an in-progress Lean 4 formalization whose proofs are not yet complete (Section 6).

Organization. Section 2 positions this work against the closest prior constructions. Section 3 fixes notation and records the (standard) monomial picture of the degree filtration. Section 4 defines the kernel basis and proves Theorem 4.5. Section 5 is the paper’s core, proving Theorem 5.2. Section 6 records the state of computational and formal verification, and Section 7 closes with directions for future work.

2. Related work

The Lin–Chung–Han novel polynomial basis. A relevant comparison is the “novel polynomial basis” of Lin, Chung, and Han [1], developed for additive Fourier transforms over characteristic-two fields. Fixing a Cantor basis of $\mathbb{F}_{2^r}$ and the associated subspace-vanishing polynomials $s_0, s_1, \dots$, their basis elements are $X_j(x) = \prod_i s_i(x)^{b_i}$ for $j = \sum_i b_i 2^i$, and satisfy $\deg X_j = j$: the basis is degree-graded, so low degree is exactly sparsity, the same relationship the monomial basis already gives for free. The kernel basis differs on every axis that matters here: it requires no characteristic assumption and no chosen subspace of a field extension, being built instead from literal power-of- X shifts; and every element has degree $N - 1$ regardless of index, so no element is itself low-degree. Theorem 5.2 exists because the kernel basis is degree-degenerate in a way the Lin–Chung–Han basis is not.

Scaled Bernstein specialization in characteristic two. There is an exact classical specialization when $\text{char } \mathbb{F} = 2$. Set $n = N - 1 = 2^m - 1$ and $j = |y|_2$. Since $X^{2^i} + 1 = (X + 1)^{2^i}$ in characteristic two,

$$K_y(X) = X^{n-j}(X + 1)^j = X^{n-j}(1 - X)^j.$$

Thus, after the reindexing $i = n - j$, the kernel family is exactly the scaled Bernstein basis $\varphi_{i,n}(X; 0, 1) = X^i(1 - X)^{n-i}$ of grade n ; see, for example, Mackey and Perović [4]. In this particular grade the usual Bernstein binomial normalization also disappears modulo two: in $\mathbb{F}_2[T]$,

$$(1 + T)^{2^m - 1} = 1 + T + \dots + T^{2^m - 1},$$

so every coefficient $\binom{2^m - 1}{j}$ is 1 modulo 2. This identifies a classical boundary case of the construction. The arbitrary-field filtration theorem below does not rely on characteristic two.

The Boolean zeta and Möbius transforms. The change of basis underlying Section 4 – summing kernel coefficients over a superset in the Boolean lattice, and inverting by an alternating sum – is the classical zeta transform and its Möbius inversion on the subset lattice, foundational to enumerative combinatorics and, under the name of the algebraic-normal-form transform, to the analysis of Boolean functions over $\mathbb{F}_2$. What is not classical, to the best of our knowledge, is the specific fact that this transform, composed with a fixed complement permutation and specialized to the maximally-degenerate kernel basis, characterizes the ordinary polynomial degree filtration via an alternating sign condition rather than a support condition. We were unable to locate this statement, or an evidently equivalent one, in the additive-FFT or coding-theory literature; Section 5 states and proves it directly.

FRI and folding-based polynomial commitments. Fast Reed–Solomon Interactive Oracle Proofs of Proximity (FRI) [5] test proximity to a low-degree Reed–Solomon codeword by repeatedly folding a polynomial via its even and odd parts, halving the degree bound at each round. BaseFold [2] generalizes this folding step to arbitrary fields via foldable codes, and FRI-Binius [3] draws an explicit connection between an additive transform built on the Lin–Chung–Han basis and binary-field FRI, prescribing concrete choices left generic in the original construction. Both lines of work establish degree-halving under folding directly, without passing through an intermediate basis-theoretic statement of the kind given here. This paper does not attempt a new folding or proximity result; the interaction of the even–odd folding operator with the kernel coefficient table is left to future work.

3. Preliminaries

Throughout, $\mathbb{F}$ denotes an arbitrary field: no assumption is made on its characteristic. Fix an integer $m \geq 1$ and set

$$N = 2^m, \quad \mathbb{B}^m = \{0, 1\}^m.$$

We write $\mathbb{F}[X]_{<N}$ for the N -dimensional $\mathbb{F}$ -vector space of polynomials of degree strictly less than N .

3.1. Binary exponent coordinates

For $a = (a_0, \dots, a_{m-1}) \in \mathbb{B}^m$, define its binary value

$$|a|_2 = \sum_{i=0}^{m-1} a_i 2^i.$$

The map $a \mapsto |a|_2$ is a bijection $\mathbb{B}^m \rightarrow \{0, \dots, N-1\}$, so every polynomial $U \in \mathbb{F}[X]_{<N}$ has a unique expansion

$$U(X) = \sum_{a \in \mathbb{B}^m} u_a X^{|a|_2},$$

a reindexing of the ordinary monomial coefficients by Boolean vectors rather than by integers, not a new representation of U .

For $a, y \in \mathbb{B}^m$ we write $y \geq a$ when $y_i \geq a_i$ for every coordinate i , and we let $\text{wt}(y) = \sum_i y_i$ denote the Hamming weight of y .

Fix $0 \leq k \leq m$ and set $q = m - k$. Every $a \in \mathbb{B}^m$ decomposes uniquely as $a = (r, s)$ with $r \in \mathbb{B}^k$ the low coordinates and $s \in \mathbb{B}^q$ the high coordinates, and

$$|a|_2 = |r|_2 + 2^k |s|_2.$$

Proposition 3.1 (Monomial low-degree condition). *Write*

$$U(X) = \sum_{(r,s) \in \mathbb{B}^k \times \mathbb{B}^q} u_{(r,s)} X^{|r|_2 + 2^k |s|_2}.$$

Then $\deg U < 2^k$ if and only if $u_{(r,s)} = 0$ for every $r \in \mathbb{B}^k$ and every nonzero $s \in \mathbb{B}^q$.

Proof. If $\deg U < 2^k$ and $s \neq 0$, then $|s|_2 \geq 1$, so $|r|_2 + 2^k|s|_2 \geq 2^k$ for every r , and the corresponding monomial cannot occur in U ; hence $u_{(r,s)} = 0$. Conversely, if $u_{(r,s)} = 0$ whenever $s \neq 0$, then $U(X) = \sum_{r \in \mathbb{B}^k} u_{(r,0)} X^{|r|_2}$ with $|r|_2 \leq 2^k - 1$ for every r , so $\deg U < 2^k$. $\square$

Proposition 3.1 is the ordinary, sparsity-based picture of the degree filtration on $\mathbb{F}[X]_{<N}$: a degree bound is equivalent to the vanishing of a set of monomial coefficients. Every degree-graded basis of $\mathbb{F}[X]_{<N}$ shares this picture. Section 4 introduces a basis for which it fails completely, and Section 5 identifies what replaces it.

4. The Boolean kernel basis

4.1. Definition and degree

Definition 4.1 (Kernel polynomial). For $y = (y_0, \dots, y_{m-1}) \in \mathbb{B}^m$, the associated *kernel polynomial* is

$$K_y(X) = \prod_{i=0}^{m-1} (X^{2^i} + y_i) \in \mathbb{F}[X].$$

Every factor $X^{2^i} + y_i$ is monic of degree 2^i .

Proposition 4.2 (Every kernel polynomial has maximal degree). *For every $y \in \mathbb{B}^m$, $\deg K_y = N - 1$, with leading coefficient 1.*

Proof. Selecting the term X^{2^i} from every factor gives the monomial $\prod_{i=0}^{m-1} X^{2^i} = X^{\sum_i 2^i} = X^{N-1}$ with coefficient 1; any other choice of terms from the factors yields a monomial of strictly lower degree, so it cannot cancel the leading term. $\square$

Proposition 4.2 is what distinguishes the kernel basis from degree-graded bases of $\mathbb{F}[X]_{<N}$, including the monomial basis and the Lin–Chung–Han novel polynomial basis discussed in Section 2: *every* kernel polynomial sits at the top of the degree range, so no element of the basis is itself low-degree. A degree bound on U can therefore not be read off as sparsity of its kernel coefficients, the way it can for a degree-graded basis (Proposition 3.1). Section 5 identifies exactly what a degree bound looks like in this basis instead.

4.2. Exact coefficient formula

For $a \in \mathbb{B}^m$, write $K_y(X) = \sum_{a \in \mathbb{B}^m} [K_y]_a X^{|a|_2}$.

Proposition 4.3 (Coefficient formula). *For every $a, y \in \mathbb{B}^m$, write $\bar{a} = (1 - a_0, \dots, 1 - a_{m-1})$ for the Boolean complement of a . Then*

$$[K_y]_a = \prod_{i: a_i=0} y_i = \begin{cases} 1, & y \geq \bar{a}, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. The monomial $X^{|a|_2}$ arises from the product by choosing X^{2^i} from the i -th factor whenever $a_i = 1$, and choosing the constant term y_i whenever $a_i = 0$; the resulting coefficient is $\prod_{i:a_i=0} y_i$. Since each $y_i \in \{0, 1\}$, this product equals 1 exactly when $y_i = 1$ for every i with $a_i = 0$. As $\bar{a}_i = 1$ exactly when $a_i = 0$ (and imposes no constraint when $\bar{a}_i = 0$), this condition is exactly $y_i \geq \bar{a}_i$ for every i , i.e. $y \geq \bar{a}$. $\square$

Consequently $\text{Supp}(K_y) = \{a \in \mathbb{B}^m : y \geq \bar{a}\}$ has size $2^{\text{wt}(y)}$: the condition $y \geq \bar{a}$ says $y_i + a_i \geq 1$ for every i , so each coordinate with $y_i = 0$ forces $a_i = 1$ on the support, while each coordinate with $y_i = 1$ leaves a_i free.

Example 4.4. For $m = 2$ ($N = 4$): $K_{(0,0)}(X) = X^3$, $K_{(1,0)}(X) = X^3 + X^2$, $K_{(0,1)}(X) = X^3 + X$, and $K_{(1,1)}(X) = X^3 + X^2 + X + 1$. The two extremal indices give $K_{0^m}(X) = X^{N-1}$ and $K_{1^m}(X) = 1 + X + \dots + X^{N-1}$, related by $(X - 1)K_{1^m}(X) = X^N - 1$.

4.3. Recursive decomposition and the basis theorem

Write $y = (b, z)$ with $b = y_0 \in \{0, 1\}$ and $z = (y_1, \dots, y_{m-1}) \in \mathbb{B}^{m-1}$, and let $K_z(Y) = \prod_{i=0}^{m-2} (Y^{2^i} + z_i)$ be the corresponding kernel polynomial for $m - 1$. Substituting $Y = X^2$ gives

$$K_{(0,z)}(X) = X K_z(X^2), \quad K_{(1,z)}(X) = (X + 1) K_z(X^2), \quad (1)$$

splitting every kernel polynomial into an even part and an odd part in X .

Theorem 4.5 (Boolean kernel basis). *The family $\{K_y(X) : y \in \mathbb{B}^m\}$ is a basis of $\mathbb{F}[X]_{<N}$.*

Proof. Both sides have dimension $N = 2^m$, so it suffices to prove linear independence, by induction on m . For $m = 1$, $K_{(0)}(X) = X$ and $K_{(1)}(X) = X + 1$ are linearly independent. Assume the claim for $m - 1$ and suppose

$$\sum_{z \in \mathbb{B}^{m-1}} \alpha_{0,z} K_{(0,z)}(X) + \sum_{z \in \mathbb{B}^{m-1}} \alpha_{1,z} K_{(1,z)}(X) = 0.$$

By (1) this reads

$$\sum_z \alpha_{1,z} K_z(X^2) + X \sum_z (\alpha_{0,z} + \alpha_{1,z}) K_z(X^2) = 0.$$

The first sum contains only even powers of X and the second only odd powers, so each must vanish separately. By the induction hypothesis the polynomials $K_z(Y)$, $z \in \mathbb{B}^{m-1}$, are linearly independent in Y , so evaluating at $Y = X^2$ preserves their independence and forces $\alpha_{1,z} = 0$ for every z ; the second identity then gives $\alpha_{0,z} = 0$ as well. $\square$

Every $U \in \mathbb{F}[X]_{<N}$ therefore has a unique kernel expansion $U(X) = \sum_{y \in \mathbb{B}^m} \lambda_y K_y(X)$; we call $\lambda : \mathbb{B}^m \rightarrow \mathbb{F}$ the *kernel coefficient table* of U . These are coordinates of U in the basis $\{K_y\}$, and should not be confused with values of U at points of $\mathbb{F}$.

4.4. The zeta transform between kernel and monomial coordinates

Comparing the two expansions of U via Proposition 4.3 gives

$$u_a = \sum_{y \geq \bar{a}} \lambda_y, \quad (2)$$

the Boolean zeta transform of λ , evaluated at the complement of a . Writing $\zeta\lambda(c) := \sum_{y \geq c} \lambda_y$ for the ordinary zeta transform of λ on the Boolean lattice, (2) says $u_a = \zeta\lambda(\bar{a})$: the monomial coefficients are the zeta transform of λ , followed by the complement permutation $a \mapsto \bar{a}$.

In matrix form $u = Z_m \lambda$ with $Z_m(a, y) = \mathbf{1}[y \geq \bar{a}]$; for $m = 1$, $Z_1 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ has $\det Z_1 = -1$, and Z_m is, up to the ordering of Boolean indices, the m -fold Kronecker power of Z_1 – hence invertible over every field. This furnishes a second, purely linear-algebraic proof of Theorem 4.5.

Ordinary Boolean Möbius inversion recovers λ from $\zeta\lambda$, $\lambda_y = \sum_{b \geq y} (-1)^{\text{wt}(b) - \text{wt}(y)} \zeta\lambda(b)$; substituting $\zeta\lambda(b) = u_{\bar{b}}$ from (2) gives

$$\lambda_y = \sum_{b \geq y} (-1)^{\text{wt}(b) - \text{wt}(y)} u_{\bar{b}}, \quad (3)$$

an identity that remains valid in characteristic 2, where addition and subtraction coincide. Both the zeta transform and Möbius inversion evaluate coordinate by coordinate on the standard butterfly schedule for the Boolean lattice, in $\frac{1}{2}N \log_2 N$ field operations each – that is, $O(N \log N)$ total for (2) or (3) together with the free $O(N)$ relabeling $a \mapsto \bar{a}$, rather than the $O(N^2)$ of a naive change of basis – confirmed empirically in Section 6.

5. The low-degree filtration

This section states and proves the paper’s main result: an exact characterization of the degree filtration of $\mathbb{F}[X]_{<N}$ directly in kernel coordinates. By Proposition 4.2, no single kernel polynomial is low-degree, so this cannot be a sparsity condition on λ the way Proposition 3.1 is a sparsity condition on u ; Theorem 5.2 identifies the condition that replaces it.

5.1. Setting and the high-coordinate sign character

Fix $0 \leq k \leq m$ and set $q = m - k$, exactly as in Section 3, but now split the *kernel* index $y \in \mathbb{B}^m$ rather than the monomial index: write $y = (x, h)$ with $x \in \mathbb{B}^k$ the low coordinates and $h \in \mathbb{B}^q$ the high coordinates, and write the kernel expansion of U accordingly,

$$U(X) = \sum_{(x, h) \in \mathbb{B}^k \times \mathbb{B}^q} \lambda_{(x, h)} K_{(x, h)}(X).$$

Definition 5.1 (High-coordinate sign character). For $h \in \mathbb{B}^q$, set $\sigma_q(h) = (-1)^{q - \text{wt}(h)}$. Equivalently $\sigma_q(h) = \prod_{j=0}^{q-1} (-1)^{1 - h_j}$: the sign flips whenever a single high coordinate is flipped. Up to the constant factor $(-1)^q$, σ_q is the Boolean character $h \mapsto (-1)^{\text{wt}(h)}$.

5.2. The main theorem

Theorem 5.2 (Low-degree filtration). *Let $U(X) = \sum_{(x,h)} \lambda_{(x,h)} K_{(x,h)}(X) \in \mathbb{F}[X]_{<N}$. Then $\deg U < 2^k$ if and only if there is a (necessarily unique) function $\mu : \mathbb{B}^k \rightarrow \mathbb{F}$ with*

$$\lambda_{(x,h)} = (-1)^{q-\text{wt}(h)} \mu_x \quad \text{for every } x \in \mathbb{B}^k, h \in \mathbb{B}^q.$$

When it exists, $\mu_x = \lambda_{(x,1_q)}$, where 1_q denotes the all-ones vector of $\mathbb{B}^q$.

5.3. Proof: degree implies the character condition

Write the monomial expansion of U along the same low/high split, as in Proposition 3.1:

$$U(X) = \sum_{(r,s)} u_{(r,s)} X^{|r|_2 + 2^k |s|_2}.$$

Splitting the zeta transform (2) along these blocks, with $\bar{r} \in \mathbb{B}^k$ and $\bar{s} \in \mathbb{B}^q$ the complements of r and s ,

$$u_{(r,s)} = \sum_{x \geq \bar{r}} \sum_{h \geq \bar{s}} \lambda_{(x,h)}. \quad (4)$$

Proof of Theorem 5.2, forward direction. Suppose $\deg U < 2^k$. By Proposition 3.1, $u_{(r,s)} = 0$ for every $r \in \mathbb{B}^k$ and every nonzero $s \in \mathbb{B}^q$. Fix such an s and, for $x \in \mathbb{B}^k$, set $F(x) = \sum_{h \geq \bar{s}} \lambda_{(x,h)}$, so by (4), $u_{(r,s)} = \sum_{x \geq \bar{r}} F(x)$ for every r . As r ranges over $\mathbb{B}^k$ so does $\bar{r}$, so $u_{(r,s)} = 0$ for every r says that the zeta transform of F on $\mathbb{B}^k$ vanishes identically; since that transform is invertible (Section 4.4), $F \equiv 0$. Thus

$$\sum_{h \geq \bar{s}} \lambda_{(x,h)} = 0 \quad \text{for every } x \in \mathbb{B}^k \text{ and every nonzero } s \in \mathbb{B}^q. \quad (5)$$

Fix x and define $G_x(b) = \sum_{h \geq b} \lambda_{(x,h)}$ for $b \in \mathbb{B}^q$. As s ranges over the nonzero vectors of $\mathbb{B}^q$, its complement $\bar{s}$ ranges over every vector of $\mathbb{B}^q$ except 1_q , so (5) says exactly $G_x(b) = 0$ for every $b \neq 1_q$. Boolean Möbius inversion recovers $\lambda_{(x,h)}$ from G_x , $\lambda_{(x,h)} = \sum_{b \geq h} (-1)^{\text{wt}(b) - \text{wt}(h)} G_x(b)$; since G_x vanishes except at $b = 1_q$, only that term survives:

$$\lambda_{(x,h)} = (-1)^{\text{wt}(1_q) - \text{wt}(h)} G_x(1_q) = (-1)^{q - \text{wt}(h)} \mu_x, \quad \mu_x := G_x(1_q) = \lambda_{(x,1_q)}.$$

□

5.4. Proof: the character condition implies low degree

Proof of Theorem 5.2, converse direction. Suppose $\lambda_{(x,h)} = (-1)^{q - \text{wt}(h)} \mu_x$ for every x, h . Let $s \in \mathbb{B}^q$ be nonzero and set $b = \bar{s}$; since $s \neq 0_q$, $b \neq 1_q$, so b has at least one zero coordinate – say $f \geq 1$ of them. For every x ,

$$\sum_{h \geq b} \lambda_{(x,h)} = \mu_x \sum_{h \geq b} (-1)^{q - \text{wt}(h)}.$$

Writing $h = b$ plus an arbitrary assignment of the f free coordinates, $\sum_{h \geq b} (-1)^{q-\text{wt}(h)} = (-1)^{q-\text{wt}(b)} \sum_{j=0}^f \binom{f}{j} (-1)^j = (-1)^{q-\text{wt}(b)} (1-1)^f = 0$ since $f \geq 1$. Hence $\sum_{h \geq s} \lambda_{(x,h)} = 0$ for every x and every nonzero s , and by (4), $u_{(r,s)} = 0$ for every r and every nonzero s . Proposition 3.1 then gives $\deg U < 2^k$. $\square$

Uniqueness of μ in Theorem 5.2 is immediate from the forward direction: μ_x is forced to equal $\lambda_{(x,1_q)}$.

5.5. Collapse of the high coordinates

For $x \in \mathbb{B}^k$, define the truncated kernel polynomial

$$K_x^{[k]}(X) = \prod_{i=0}^{k-1} (X^{2^i} + x_i),$$

the degree- $(2^k - 1)$ kernel polynomial for $m = k$ built from only the low coordinates.

Proposition 5.3 (Collapse formula). *If $\deg U < 2^k$, then $U(X) = \sum_{x \in \mathbb{B}^k} \mu_x K_x^{[k]}(X)$, with μ as in Theorem 5.2.*

Proof. By Theorem 5.2, $U(X) = \sum_x \mu_x \sum_{h \in \mathbb{B}^q} (-1)^{q-\text{wt}(h)} K_{(x,h)}(X)$. Since $K_{(x,h)}(X) = K_x^{[k]}(X) \cdot \prod_{j=0}^{q-1} (X^{2^{k+j}} + h_j)$, and $(-1)^{q-\text{wt}(h)} = \prod_{j=0}^{q-1} (-1)^{1-h_j}$ by Definition 5.1, the inner sum factors coordinate by coordinate:

$$\sum_{h \in \mathbb{B}^q} (-1)^{q-\text{wt}(h)} K_{(x,h)}(X) = K_x^{[k]}(X) \prod_{j=0}^{q-1} \left[\sum_{h_j \in \{0,1\}} (-1)^{1-h_j} (X^{2^{k+j}} + h_j) \right].$$

Each bracketed factor equals $-(X^{2^{k+j}} + 0) + (X^{2^{k+j}} + 1) = 1$, so the product is $K_x^{[k]}(X)$ and $U(X) = \sum_x \mu_x K_x^{[k]}(X)$. $\square$

5.6. The character-side low-degree space

Definition 5.4.

$$CH_{m,k} = \left\{ \lambda \in \mathbb{F}^{\mathbb{B}^m} : \lambda_{(x,h)} = (-1)^{q-\text{wt}(h)} \mu_x \text{ for some } \mu : \mathbb{B}^k \rightarrow \mathbb{F} \right\}.$$

Corollary 5.5. $\deg U < 2^k \iff \lambda \in CH_{m,k}$, and $\dim CH_{m,k} = 2^k$.

Proof. The equivalence is Theorem 5.2. The map $\mu \mapsto \lambda$ of Definition 5.4 is linear and, by the uniqueness clause of Theorem 5.2, injective, with image exactly $CH_{m,k}$; hence $\dim CH_{m,k} = \dim \mathbb{F}^{\mathbb{B}^k} = 2^k$, matching $\dim \mathbb{F}[X]_{<2^k} = 2^k$. $\square$

h	$\sigma_2(h)$	$\lambda_{(x,h)}$
$(0,0)$	1	μ_x
$(1,0)$	-1	$-\mu_x$
$(0,1)$	-1	$-\mu_x$
$(1,1)$	1	μ_x

Table 1: The character condition for $q = 2$.

5.7. Worked example: $m = 3$, $k = 1$

Here $q = 2$. Table 1 lists the character value $\sigma_2(h) = (-1)^{2-\text{wt}(h)}$ for each $h \in \mathbb{B}^2$ and the resulting kernel coefficient in the fiber over a fixed x .

Figure 1 shows the same data laid out on the Boolean square $\mathbb{B}^2$: an edge flips one coordinate of h , and the sign flips with it, so the character condition colors $\mathbb{B}^2$ like a checkerboard, μ_x on one part and $-\mu_x$ on the other. The same holds on $\mathbb{B}^q$ for any q : adjacent vertices (differing in one coordinate) always carry opposite signs, since flipping a single coordinate of h changes $\text{wt}(h)$ by exactly 1.

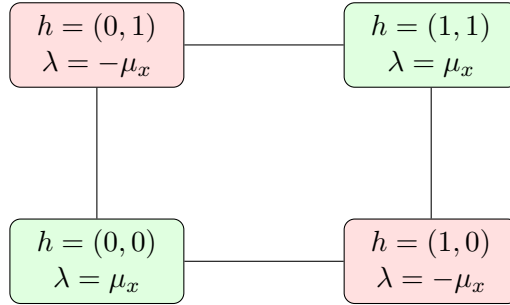


Figure 1: The character condition of Table 1 as a checkerboard coloring of $\mathbb{B}^2$: each edge changes one coordinate of h and flips the sign.

By Proposition 5.3, $U(X) = \mu_0 K_{(0)}^{[1]}(X) + \mu_1 K_{(1)}^{[1]}(X)$; since $K_{(0)}^{[1]}(X) = X$ and $K_{(1)}^{[1]}(X) = X + 1$ (as in Example 4.4, restricted to $k = 1$), this is $U(X) = \mu_0 X + \mu_1(X + 1)$, of degree less than 2 as required.

5.8. Boundary cases

When $k = m$, $q = 0$: there is no high block, the character condition is vacuous, and $CH_{m,m} = \mathbb{F}^{\mathbb{B}^m}$ places no restriction on λ – consistent with $\deg U < 2^m$ holding for every $U \in \mathbb{F}[X]_{<N}$. When $k = 0$, $q = m$: μ is a single scalar and the character condition forces the full alternating pattern $\lambda_h = (-1)^{m-\text{wt}(h)}\mu$ across all of $\mathbb{B}^m$; the corresponding U is the constant polynomial μ .

5.9. Characteristic two

Theorem 5.2 makes no assumption on the characteristic of $\mathbb{F}$, so it applies unchanged when $\mathbb{F}$ has characteristic 2 – the setting of the binary tower fields underlying Binius and

FRI-Binius [3]. In that case the theorem simplifies further: since $-1 = 1$, the sign $(-1)^{q-\text{wt}(h)}$ is identically 1 regardless of h , so the checkerboard coloring of Figure 1 collapses to a single color. The character condition then reads simply

$$\deg U < 2^k \iff \lambda_{(x,h)} \text{ is independent of } h,$$

i.e. constant across every high-coordinate fiber – a constancy condition rather than an alternating one. The alternating-sign phenomenon is thus specific to characteristic $\neq 2$; over $\mathbb{F}_2$, the same theorem reduces to an ordinary invariance statement.

5.10. Summary

The Boolean kernel basis converts an ordinary degree bound into an exact character condition:

$$\deg U < 2^k \iff \lambda_{(x,h)} = (-1)^{q-\text{wt}(h)} \mu_x \text{ for some } \mu : \mathbb{B}^k \rightarrow \mathbb{F}.$$

Section 6 records the computational and formal checks of this theorem. A natural direction for future work is to study how the even–odd folding step of an FRI-style protocol acts on the kernel coefficient table and interacts with the degree filtration characterized above.

6. Formal verification

Two distinct verification layers accompany Theorem 4.5 and Theorem 5.2. First, a committed Rust implementation over the Goldilocks prime field provides a reproducible computational cross-check. It computes the zeta transform and its Möbius inversion both by a direct $O(N^2)$ reference method and by the fast $O(N \log N)$ butterfly schedule, asserts agreement between the two implementations on the tested range, and checks both directions of Theorem 5.2 for every k up to $m = 8$.

Second, a Lean 4 formalization against Mathlib is in progress. At the submission stage its central proofs are not yet complete, so the Lean artifact is reported as formalization work in progress rather than as a machine-checked proof of this paper. The mathematical arguments in the present article are self-contained and do not depend on either computational artifact.

7. Conclusion and future work

We showed that the kernel polynomials $K_y(X) = \prod_i (X^{2^i} + y_i)$ form a basis of $\mathbb{F}[X]_{<N}$ over an arbitrary field in which every element has maximal degree, and that this degeneracy is exactly compensated by an alternating sign condition characterizing the ordinary degree filtration: $\deg U < 2^k$ if and only if $\lambda_{(x,h)} = (-1)^{q-\text{wt}(h)} \mu_x$. The proof runs through the Boolean zeta transform and its Möbius inversion, gives an explicit collapse formula recovering a low-degree polynomial from its free coordinates, and identifies the resulting 2^k -dimensional character space $CH_{m,k}$. Section 6 records the committed Rust cross-check and the clearly separated status of the ongoing Lean 4 formalization.

This paper is deliberately narrow. A natural next question is how the even–odd folding operator used in FRI-style proximity testing acts on the kernel coefficient table and on

the spaces $CH_{m,k}$. That question is outside the scope of the present article: no folding or proximity theorem is claimed here. The filtration theorem provides an algebraic starting point for such later work.

Code and data availability

The manuscript source, Rust implementation, tests, benchmark harness, and the in-progress Lean 4 formalization are publicly available at [public GitHub repository](#). No archival DOI is claimed for this version.

Declaration of competing interest

The author declares no competing financial interests.

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